ivcheck: Tests of Instrumental Variable Validity in R

by

Abstract The **ivcheck** package implements three falsification tests of the identifying assumptions behind instrumental variable estimation of the local average treatment effect: Kitagawa (2015), Mourifie and Wan (2017), and Frandsen, Lefgren, and Leslie (2023). Each test had existed only as author-hosted replication code in Stata or Matlab prior to this package. **ivcheck** exposes each as a named R function with S3 methods for fitted **fixest** and **ivreg** models, plus a one-shot wrapper that runs every applicable test on a fitted model in a single call. The implementation uses the variance- weighted Kolmogorov-Smirnov form of Kitagawa, the full Chernozhukov- Lee-Rosen intersection-bounds inference with adaptive moment selection for Mourifie-Wan, and the asymptotic chi-squared form of Frandsen-Lefgren-Leslie with multivalued-treatment support. Monte Carlo validation confirms that the empirical null distributions match the published asymptotic reference distributions.

1 Introduction

The **ivcheck** package provides falsification tests of the identifying assumptions behind instrumental variable (IV) estimation of the local average treatment effect (LATE). It implements three tests from the econometric literature (Kitagawa, 2015; Mourifié and Wan, 2017; Frandsen et al., 2023), each with S3 methods for fitted **fixest** and **ivreg** model objects, plus a wrapper that runs every applicable test on a single fitted model.

Applied IV practice has a tooling gap. The identifying assumptions of the Imbens and Angrist (1994) LATE framework, the local exclusion restriction and monotonicity, are routinely defended by hand-wave rather than tested, despite each having testable implications derived in the methodological literature since 2015. The reason is not conviction but distribution: Kitagawa's (2015) test was released with supplementary Matlab code on the author's website; Mourifie and Wan's (2017) reformulation relies on the Chernozhukov-Lee-Rosen (2013) `clrtest` Stata module; Frandsen, Lefgren, and Leslie (2023) ship a Stata SSC module `testjfe`. None of the three has existed as a maintained R package until now. The adjacent **ivDiag** (Lal et al., 2024) covers weak-instrument diagnostics (effective F, Anderson-Rubin, valid-t, local-to-zero) but does not implement LATE-validity tests; the R ecosystem has lacked a counterpart to that package for the falsification-test family.

ivcheck closes that gap. The package has three functional commitments. First, each test is a named R function (`iv_kitagawa`, `iv_mw`, `iv_testjfe`) that accepts either raw `(y, d, z)` vectors or a fitted IV model through S3 dispatch. Second, the implementations are faithful to the published forms: Kitagawa's variance-weighted Kolmogorov-Smirnov statistic from section 4 of the paper, the full Chernozhukov-Lee-Rosen intersection-bounds inference with Andrews-Soares (2010) adaptive moment selection for the conditional Mourifie-Wan test, and the asymptotic chi-squared form of Frandsen-Lefgren-Leslie extended to multivalued treatment per section 4 of that paper. Third, a one-shot `iv_check` wrapper detects which tests are applicable from the structure of a fitted IV model and runs them, producing a tidy summary ready for inclusion in a paper's appendix.

2 Background

2.1 The LATE framework and its identifying assumptions

The Imbens-Angrist (1994) LATE framework identifies the causal effect of a binary treatment `D` on an outcome `Y` for the subpopulation of compliers under two untestable-looking assumptions about an instrument `Z`: the local exclusion restriction (`Z` affects `Y` only through `D`) and monotonicity (no unit's treatment assignment flips in the opposite direction to the average response to `Z`). Together with independence, these identify the LATE.

2.2 Testable implications

Under LATE identification, the joint distribution of `(Y, D, Z)` is constrained. Kitagawa (2015) shows that the population conditional joint CDFs `F(y, d | z) = Pr(Y <= y, D = d | Z = z)` must satisfy a set of stochastic dominance inequalities indexed by `(y, d)`, and that these inequalities are sharp characterisations of the identifying assumptions. Mourifié and Wan (2017) reformulates the implica-

tions as conditional moment inequalities that can accept covariates `X` through the Chernozhukov-Lee-Rosen (2013) intersection-bounds framework. Frandsen et al. (2023) derive a dedicated test for the judge-fixed-effects design where the instrument is a set of mutually exclusive dummies: under the joint null, the per-judge outcome mean `mu_j = E[Y | J = j]` is a linear function of the per-judge propensity vector `P_j`.

All three tests share a structural feature: they test necessary conditions, not sufficient ones. Failure to reject is evidence of no detectable violation at level `alpha`; rejection is evidence that at least one of exclusion or monotonicity has failed. The tests do not localise which assumption failed.

3 Package design

3.1 Architecture and dependencies

ivcheck is a pure-computation package. No network access is performed at runtime. R version 4.1.0 or later is required. Hard imports are `cli`, `stats`, and `parallel`; the bootstrap uses `parallel::mclapply` on POSIX systems with an explicit 2-core cap under R's `_R_CHECK_LIMIT_CORES_` environment variable. **fixest**, **ivreg**, **modelsummary**, and **broom** are in Suggests and are conditionally registered at load time via `.onLoad`.

3.2 The `iv_test` class

Every test returns an object of class `iv_test`, a list with uniform slots: the test statistic, the bootstrap or asymptotic p-value, the significance level, the bootstrap statistics vector, a `binding` list identifying the `(z, z', d, y)` configuration of the observed statistic, and the sample size. Test-specific slots carry additional information: `weighting` for `iv_kitagawa`, `conditional` and `kappa_n` for `iv_mw`, and `pairwise_late` and `worst_pair` for `iv_testjfe`. A `print.iv_test` method emits a three-line summary; `plot.iv_test` shows the bootstrap distribution with the observed statistic marked.

3.3 S3 dispatch on fitted IV models

Each test function is an S3 generic with methods for `fixest`, `ivreg`, and the default numeric-vector interface. The `fixest` method recovers `y` via `model.matrix(m, type = "lhs")`, `d` via `fitted(fs) + residuals(fs)` from the first-stage fit, and `z` from the first-stage design matrix by column name; no reliance on the call environment, so the extractor works after the model goes out of scope. The `ivreg` method uses `object$y`, `object$endogenous`, and `object$instruments` directly. Both error cleanly on non-IV models and on models with more than one endogenous variable.

3.4 The `iv_check` wrapper

`iv_check` inspects the model structure, detects which tests are applicable (binary versus discrete treatment, number of instrument levels, presence of covariates), and runs them. The return is an `iv_check` object containing a tidy table with one row per test (name, statistic, p-value, verdict) plus an overall verdict string. When a user requests a specific test that is not applicable, `iv_check` warns and skips rather than silently dropping the request.

3.5 Bootstrap machinery

All three tests share a multiplier-bootstrap core. Rademacher weights are the default; a Mammen two-point option is available. The bootstrap process for each test is a linear transformation of the centred indicator or residual process, computed once per bootstrap replication via matrix-vector operations. Cell cap of 2 cores under `_R_CHECK_LIMIT_CORES_` keeps the package within CRAN example policy; interactive use defaults to `min(4, detectCores() - 1)`.

4 The Kitagawa (2015) test

4.1 Statistic and bootstrap

`iv_kitagawa` implements the variance-weighted Kolmogorov-Smirnov statistic of Kitagawa (2015) equation 2.1. For each pair `(z_L, z_H)` of instrument levels pre-ordered by first-stage `E_hat[D | Z]`,

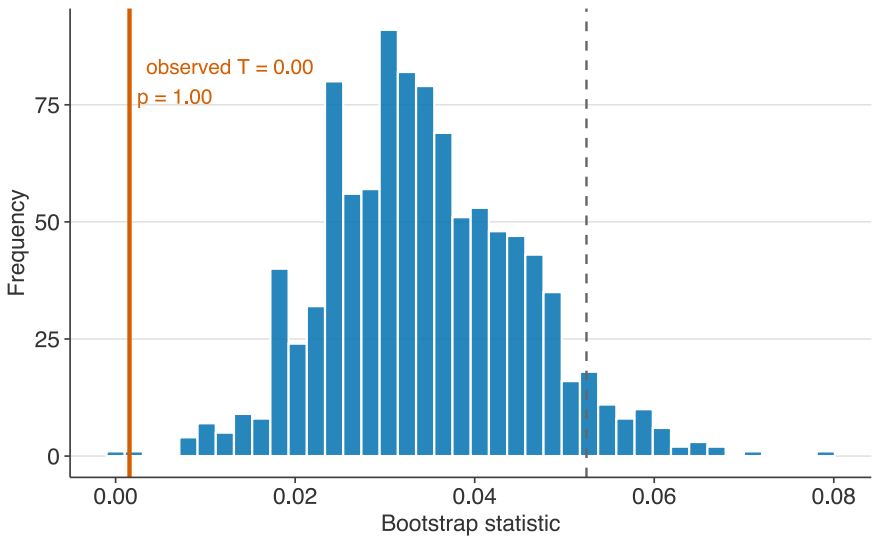

Figure 1: Kitagawa (2015) bootstrap distribution for the Card (1995) proximity-to-college IV, N = 2991. Observed test statistic (solid line) and the 95th percentile of the multiplier bootstrap (dashed). Treatment is a binary completed-college indicator (`educ >= 16`); the instrument is `near_college`. The observed statistic lies in the upper tail of the bootstrap distribution; the interval-sup test rejects the binary-discretised IV's testable implications at the 5% level. This should not be interpreted as a rejection of Card's original IV, which targets continuous years of schooling. The binary threshold at `educ >= 16` creates mixed complier subpopulations that make the testable implications bite.

and for each `d in {0, 1}`, the testable null implies that the interval-probability difference `P([y, y'], d | Z = z_L) - P([y, y'], d | Z = z_H)` is non-positive for every `y <= y'` (with the sign reversed for `d = 0`). Kitagawa's studentised statistic is

$$T_n = \sqrt{\frac{n_L n_H}{n}} \max_{(z_L, z_H, d)} \sup_{y \le y'} \frac{\left[\Delta F_{\text{hat}}(y, y'; d, z_L, z_H)\right]^+}{\hat{\sigma}(y, y'; d, z_L, z_H) \vee \xi}, \tag{1}$$

where `Delta F_hat` is the signed interval-probability difference, `sigma_hat` is the plug-in binomial-mixture standard error with mixture weights matching Kitagawa's equation 2.1, and `xi` is a small positive trimming constant that floors the denominator on intervals with vanishing estimated variance. Pairs are pre-ordered so the inequalities are one-sided and the statistic is strictly non-negative. Critical values come from a multiplier bootstrap: for each replication, resampled subsamples from the pooled empirical distribution produce a bootstrap analogue of `Delta F_hat` normalised by the same data-derived standard error. The bootstrap p-value is the fraction of replications whose statistic exceeds `T_n`.

4.2 Implementation

The supremum over `(y, y')` is computed on a quantile grid of observed outcomes (default 50 points), which is equivalent to evaluation at every sample-point pair up to Monte Carlo error under Kitagawa's Theorem 2.1. The `weighting` argument selects the variance-weighted form of equation 2.1 (default) or the unweighted form of equation 2.2 (`weighting = "unweighted"`). The two are asymptotically equivalent under the null boundary; the weighted form has better finite-sample power when instrument cells have unequal sizes.

4.3 Card (1995) illustration

The bundled `card1995` dataset is a cleaned extract of 2,991 observations from the Card (1995) proximity-to-college IV for returns to schooling, with a binary `college` indicator added (`educ >= 16`). Figure 1 shows the bootstrap distribution of the Kitagawa statistic on this data using the `college` indicator as the binary treatment and `near_college` as the instrument.

5 The Mourifie-Wan (2017) test

5.1 Series regression with adaptive moment selection

For unconditional use without covariates, `iv_mw` delegates to the Kitagawa core with variance weighting; the two tests are asymptotically equivalent in that case and agree numerically under identical random seeds. The non-trivial implementation is the conditional path.

With a covariate vector `x`, `iv_mw` estimates the conditional CDF `F(y, d | X = x, Z = z)` by cubic-polynomial series regression of the indicator `1{Y <= y, D = d}` on a basis of `X`, restricted to observations at each level of `Z`. Standard errors are the heteroscedasticity-consistent sandwich `b(x)' (B'B)^{-1} (B' \text{diag}(r^2) B) (B'B)^{-1} b(x)` where `b(x)` is the basis evaluated at the target `x`. The studentised positive-part statistic is supped over a grid of `(y, x)` quantile points.

Critical values come from a multiplier bootstrap with Andrews-Soares (2010) adaptive moment selection. Let `xi_hat_k` denote the observed studentised statistic at index `k`. The bootstrap sup is restricted to the contact set `S = { k : xi_hat_k >= -kappa_n }` where `kappa_n = sqrt(log(log(n)))`. Moments with `xi_hat_k` well below the boundary are treated as strictly non-binding and dropped, giving tighter critical values than the plug-in least-favourable path without compromising size.

5.2 Implementation surface

The user-facing arguments control the grid density (`x_grid_size`, `y_grid_size`), the basis (`basis_order`), and whether adaptive selection is applied (`adaptive = TRUE` by default). Bootstrap defaults (`n_boot = 1000`) deliver publication-grade p-values at `alpha = 0.05`.

6 The Frandsen-Lefgren-Leslie (2023) test

6.1 Statistic for binary treatment

`iv_testjfe` implements the asymptotic form of the Frandsen-Lefgren-Leslie (2023) test for designs with a binary treatment and an instrument consisting of mutually exclusive dummy variables (the leniency-of-assigned-judge design). Under the joint null, the Wald estimator `(mu_j - mu_k) / (p_j - p_k)` identifies the same complier LATE for every pair of judges `(j, k)`, where `mu_j = E[Y | J = j]` and `p_j = E[D | J = j]`. Under binary treatment, the overidentification test that all pairwise LATEs agree is algebraically the weighted-least-squares test that `mu_j = alpha + beta * p_j` holds, divided by a pooled structural-residual variance estimator:

$$T_n = \frac{\sum_j n_j \left(\mu_j - \hat{\alpha} - \hat{\beta}\, p_j\right)^2}{\hat{\sigma}^2}, \qquad \hat{\sigma}^2 = \frac{1}{n-K} \sum_j \sum_{i:J_i=j} (y_i - \hat{\alpha} - \hat{\beta} d_i - \bar{u}_j)^2, \tag{2}$$

where `bar u_j` is the within-judge mean of the structural residuals and the reference distribution is `chi^2_{K - 2}`. Using structural residuals rather than within-judge variance of `y` removes the first-stage binomial contribution of `D`, which otherwise inflates `sigma^2` by approximately `1 + p(1 - p)` relative to the correct denominator.

The `method = "bootstrap"` option replaces the chi-squared reference with a multiplier bootstrap of the restricted-residual process for small-`K` robustness. This is a simplification of the full Frandsen et al. (2023) test, which uses a flexible polynomial or spline specification for the outcome-propensity relationship and the Andrews-Soares (2010) moment-inequality procedure for the bounded-slope restriction. The chi-squared linearity test implemented here is the special case of their asymptotic form with `phi(p) = alpha + beta * p` (linear specification) and without the bounded-slope constraint.

6.2 Multivalued treatment

Section 4 of Frandsen, Lefgren, and Leslie (2023) extends the test to multivalued `D`. For `D` with `M + 1` distinct values, the scalar propensity `p_j` becomes an `M`-vector, the fit becomes a multiple WLS regression of `mu_j` on `(P(D = 1 | J), ..., P(D = M | J))`, and the reference distribution is `chi^2_{K - M - 1}`. `iv_testjfe` supports both binary and multivalued treatments; the extension requires no additional user input.

The Kitagawa (2015) and Mourifie-Wan (2017) tests have an analogous extension to multivalued treatment via Sun (2023), who generalises the stochastic-dominance inequalities to cumulative-tail events `P(Y <= y, D <= d | Z)` and `P(Y <= y, D >= d | Z)`. `iv_kitagawa` implements this extension

transparently: when the treatment vector has more than two distinct values, the test label switches from "Kitagawa (2015)" to "Sun (2023)" and the statistic aggregates violations across `2M` inequality families instead of `2`. Sun's asymptotic validity and bootstrap procedure carry through from the binary case.

6.3 Diagnostic output

The returned object includes the `K x K` matrix `pairwise_late` of pairwise Wald estimates (binary case only), and `worst_pair`, the judge pair whose Wald LATE deviates most from the fitted slope. These are diagnostic in the sense of the paper's figures: a pair whose Wald LATE is far from the common slope is the first place to look when investigating a rejection.

7 Case study: Card (1995) proximity to a four-year college

The Card (1995) IV estimate of the return to schooling is the canonical applied IV illustration. Report, on the same data, both the IV estimate and the **ivcheck** falsification tests:

```
library(fixest)
library(ivcheck)

data(card1995)
m <- feols(
  lwage ~ age + married + black + south | college ~ near_college,
  data = card1995
)
iv_check(m, n_boot = 1000)
```

The wrapper detects that `college` is binary and `near_college` is a binary instrument, runs Kitagawa and Mourifie-Wan, and returns the tidy diagnostic table. The Kitagawa interval-sup statistic rejects at the 5% level on this binary discretisation of Card's continuous schooling variable: a finding about the binary form of the IV question, not the original continuous-D one. Users running the test on their own binary-IV designs should inspect the `binding` element of the result to see which outcome interval carries the violation, and consider whether the discretisation itself is driving the finding.

8 Monte Carlo validation

8.1 Null distribution of `iv_testjfe`

Figure 2 shows the empirical distribution of `T_n` from equation (2) over 300 Monte Carlo replications under a valid-IV null, overlaid with the asymptotic chi-squared density with `K - 2 = 18` degrees of freedom. Empirical mean is 18.0 against target 18.0; empirical variance is 35.1 against target 36.0; the 95th percentile is 29.4 against the chi-squared critical value of 28.9. Empirical size at nominal 5% is 6.5%, in line with sampling error at this Monte Carlo budget.

8.2 Power under exclusion violation

Figure 3 shows the empirical rejection rate of `iv_kitagawa` under a D-specific direct effect of `Z` on `Y`, the class of exclusion violation the Kolmogorov-Smirnov test is designed to detect. At `delta = 0` (null) the empirical size is roughly nominal; by `delta = 1.5` in units of `sigma`, power approaches one.

8.3 Judge-IV diagnostic figures

Figure 4 plots per-judge outcome means against per-judge propensities for a valid-IV simulation and an exclusion violation. Under validity the points fit the regression line; under a violation they deviate in a pattern related to the direct judge effect.

Figure 5 shows the pairwise Wald LATE matrix for the violation design. Under validity, every off-diagonal entry estimates the common complier LATE; under the violation, pairs involving specific judges diverge markedly. `iv_testjfe()$worst_pair` surfaces the pair with the largest deviation from the fitted slope.

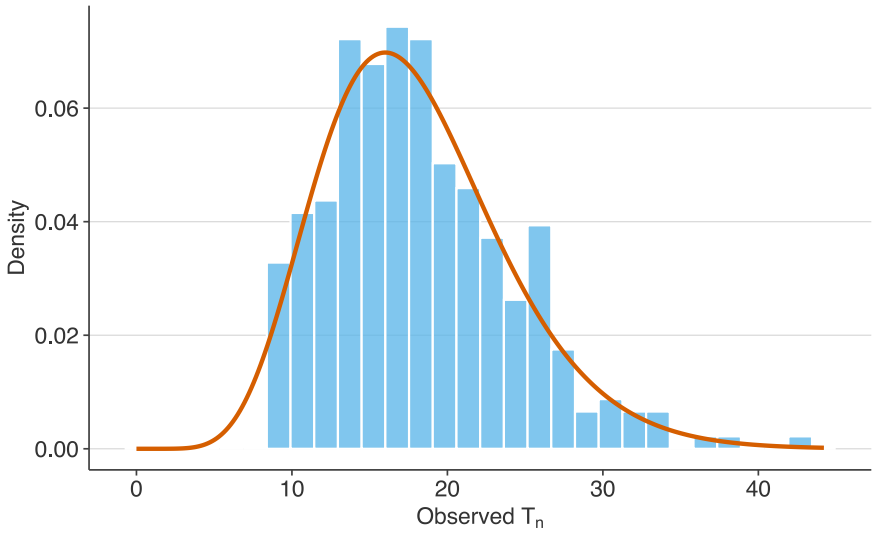

Figure 2: Monte Carlo null distribution of the Frandsen-Lefgren-Leslie test statistic, K = 20 judges and N = 3000 observations per replication. Histogram shows the empirical density from 300 replications under the null (Y independent of J given D). Red curve is the asymptotic chi-squared distribution with K - 2 = 18 degrees of freedom implied by equation (2). Agreement is close in mean, variance, and upper-tail quantile.

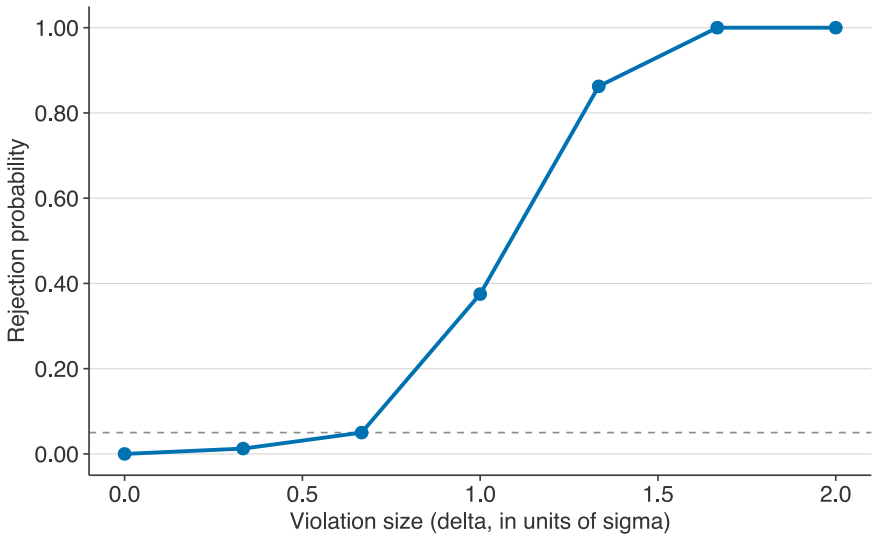

Figure 3: Empirical power of `iv_kitagawa` under a D-specific direct-effect exclusion violation, N = 500 per replication. Horizontal dashed line shows the nominal 5% level. The simulator generates data with $Y = D + \delta\sigma \cdot D \cdot (Z - z_L) + \varepsilon$, directly violating the testable inequality for the $d = 1$ cells. Power approaches one by $\delta = 1.5\sigma$ with 80 Monte Carlo replications per grid point.

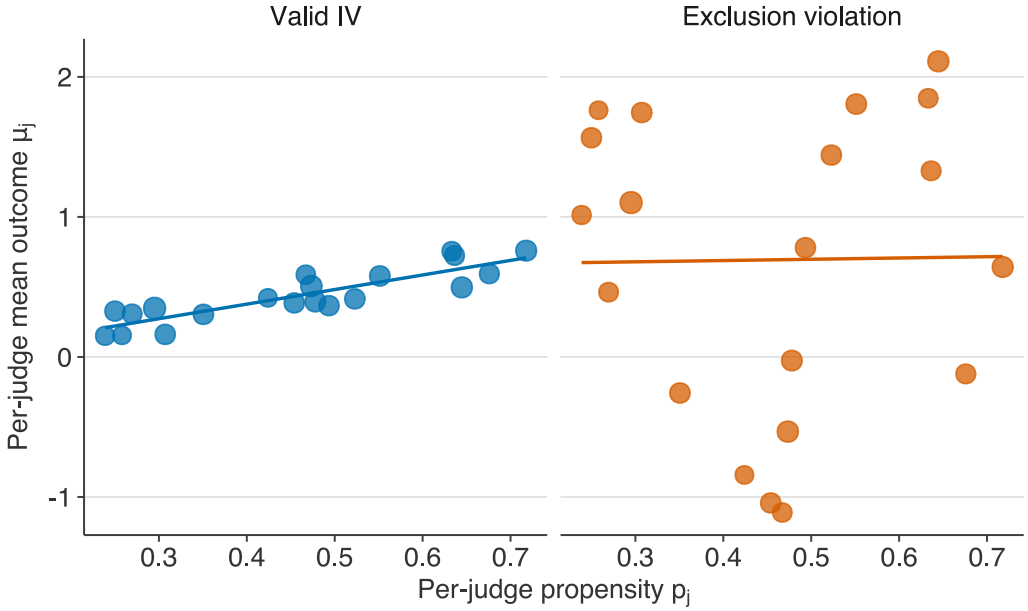

Figure 4: Per-judge outcome means versus per-judge treatment propensities, K = 20 judges and N = 3000. Left: valid-IV null ($Y = D + \varepsilon$). Right: exclusion violation with direct judge effect $1.5\sin(0.5J)$. Point size is proportional to judge-cell sample size; lines are weighted-LS fits. The violation design produces a systematic departure from linearity that `iv_testjfe` detects at $p < 10^{-4}$.

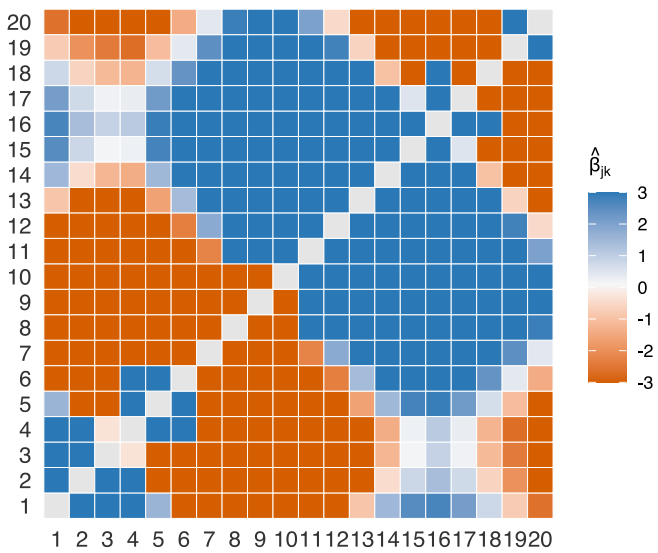

Figure 5: Pairwise Wald LATE matrix $(\mu_j - \mu_k)/(p_j - p_k)$ under the violation design from Figure 4. Entries are colour-shaded around the fitted slope; deviations in either direction signal pairs whose Wald estimates are inconsistent with a common complier LATE. The matrix is returned as `pairwise_late` on the `iv_test` object; the worst-offender pair is also returned as `worst_pair` for direct inspection.

9 Limitations

Five limitations apply.

1. **ivcheck** is a testing package, not an estimator. Use **fixest** or **ivreg** for the IV estimation itself; **ivcheck** operates on the fitted model or on raw vectors.
2. Version 0.1.0 supports discrete instruments only. Continuous `Z` must be discretised into bins (quartiles or quintiles) before passing to `iv_kitagawa` or `iv_mw`, at modest cost in power. A nonparametric continuous-Z extension is on the v0.2.0 roadmap.
3. Fuzzy regression discontinuity has a dedicated test (Arai et al., 2022) that is not yet in **ivcheck**. Its different infrastructure (running variable, cutoff, local-polynomial bandwidth, bias correction) does not fit the current spine of S3 dispatch on fitted IV models, so a dedicated `iv_frd()` function is deferred to v0.2.0.
4. `iv_testjfe` implements the asymptotic chi-squared linearity test that corresponds to the Frandsen et al. (2023) statistic under a linear outcome-propensity specification. The full flexible-series test with the bounded-slope moment-inequality restriction is planned for v0.2.0. Users needing the published test with the flexible basis should run Frandsen's Stata `testjfe` module in the interim.
5. Rejection of any of the three tests is evidence of violation but does not localise the failed assumption. Non-rejection is evidence of no detectable violation at level `alpha`, not proof of validity. Report non-rejection as such.

10 Conclusion

ivcheck closes a tooling gap between the econometric literature on LATE falsification and applied IV practice. The package is available on CRAN; source and issue tracker at `https://github.com/charlescoverdale/ivcheck`.

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